Basslines, Bodies, and Beehive Haircuts: What do we think about when we listen to music?

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The data that support the findings of this study are openly available at: https://osf.io/aepy3

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Abstract

Music influences distinct and specific mental content, including fictional narratives, autobiographical recall, and visual mental imagery. These imaginative responses are relevant in specialised contexts like therapy and play a vital role in everyday cognition, creativity, memory, attention, and well-being. However, prior studies are typically constrained to one isolated thought type, use limited musical diversity, or interpret intricate imaginings through quantitative ratings—flattening the complexity of these experiences. Computational qualitative analyses can process large datasets, but fail to capture contextual richness and nuance available through human-driven qualitative analysis. What is missing is an open, manual qualitative examination of the fuller breadth of music-influenced mental content (MIMC)—an approach capable of preserving the depth and human subtleties that quantitative, predefined, isolated, and computational approaches overlook. We analysed participants′ free-response MIMC descriptions, not restricting probing to isolated types or predefined thought-type categories. Music stimuli—120 instrumental clips from the MUSIFEAST-17 stimulus set—spanned 10 stylistically diverse genres: Ambient, Classical, Electronic, Film, Funk, Jazz, Metal, Pop, 60s-pop, 80s-pop. A total of 1,786 UK and US participants aged 18–75 years listened to 10 clips (one from each genre), reported whether they experienced imaginative responses, and provided open-text descriptions. Rigorous, mixed-method qualitative analysis was conducted on 14,091 MIMC descriptions, involving multiple coding passes, multi-coder reliability checks, thematic analysis, and co-occurrence network analysis. The resulting thematic framework functions as a typology of music-influenced "thoughtscapes", revealing multi-faceted MIMC unfolding across narrational, referential, and experiential dimensions, offering a structured account of a fuller spectrum of listeners′ imaginative experiences.

Key words: Music-influenced mental content, Imagination, Qualitative, Mixed-method, Typology

Introduction

Everyday music and imagination

We weave music into the fabric of our everyday lives, from bars, concerts, films, and video games, to gyms, commutes, hold-lines, and home kitchens. Whether it sits in the background of routine activities or commands our full attention, music influences a wide range of psychological responses, extending through affect and aesthetic evaluations into diverse and content-rich imagination (Eerola & Vuoskoski, 2012; Jakubowski et al., 2024; Nieminen et al., 2011; van der Walle et al., 2025b). To capture this range of often co-occurring music-related mental experiences within one unified conceptual framework, we introduce the term *Music-Influenced Mental Content* (MIMC), defined here as any mental content people describe as being shaped by, influenced by, attributed to, or associated with music stimuli and its listening context, without requiring an explicit causal claim (e.g. "evoked", "induced"). Framed this way, MIMC captures a heterogeneous range of multimodal experiences and unfolding scenes, spanning visual mental imagery (VMI), imagined narratives, memories, and emotional or musical appraisals (Dahl et al., 2023; Deroy, 2020; Gabrielsson, 2011; Jakubowski & Ghosh, 2021; Margulis, 2017; Taruffi & Küssner, 2019), and provides a common language for investigating how such content emerges, interacts, and evolves over time.

Researching how music shapes such our thoughts has wide-reaching implications. Because music is encountered in contexts of both active engagement (e.g. therapy, deliberate listening) and passive exposure (e.g. workplaces, retail spaces, social environments), understanding its capacity to stimulate imaginative thought is relevant to both specialised applications as well as how music shapes everyday cognition. These imaginative responses—central to our ability to reflect, create, and connect—play a vital role in attention, memory, social interaction, and well-being (Forgeard & Kaufman, 2016). Hence, insights from this line of work extend across disciplines, from education and healthcare to the creative industries and consumer engagement. Moreover, music's temporally unfolding nature offers a unique lens into how imaginative thought develops dynamically in response to shifting acoustic features and listener associations (contextual, emotional, semantic) (Margulis et al., 2022a; Margulis & Jakubowski, 2024). Research on MIMC, therefore, provides a powerful framework for investigating imagination in action. Crucially, music's reliable capacity to influence richly varied imaginative content offers a particularly valuable tool for probing the contents of everyday thought.

Current framework of music-influence mental content

Previous studies on MIMC have typically examined specific thought types in isolation (e.g. directed narrative imaginings (Herff et al., 2021), memories (Jakubowski & Ghosh, 2021), visual imagery (Belfi, 2019; Presicce & Bailes, 2019)) and with a limited amount of music and stylistic diversity. This approach has left gaps in our understanding of interactions between different thought types, the effects of varying musical style, and the nuance within these imaginative experiences. Moreover, features of these thoughts are often assessed via ratings including, for instance, the emotionality, vividness, or spontaneity of thoughts (Jakubowski & Francini, 2023; Margulis, 2017; Presicce & Bailes, 2019), rather than in-depth analysis of their content. By constraining responses to quantitative scales and predefined categories of thought (Jakubowski et al., 2024; van der Walle et al., 2025b), the variety and complexity of possible thoughts that could be experienced risks being restricted and flattened.

A more open approach focuses on the *qualitative contents* of MIMC. Allowing participants to freely describe their thoughts from music listening enables a deeper and more comprehensive view of the MIMC experience. Some studies have analysed these open-text responses with automated computational methods, such as Natural Language Processing (NLP), which capture word frequencies, thematic categories, or semantic similarities (Jakubowski and Ghosh, 2021; Margulis et al., 2022b; Taruffi et al., 2023). While automated methods can process large datasets and offer quantitative objectivity, they cannot fully capture the nuanced and context-rich details of manual, human-driven qualitative analyses. Manual coding instead allows for the discovery of subtle patterns, deeper themes, emergent ideas, and richer contextual meanings that can't easily be captured by statistical analysis.

To date, only a small number of studies have applied manual qualitative methods to MIMC, and most have focused narrowly on VMI (Dahl et al., 2023; Küssner & Eerola, 2019; Hashim et al., 2023). These studies revealed similar, prominent categories such as "settings"/"landscape or scene", "characters"/"human"/"people", "action"/"movements", "narratives"/"events", "memories"/"time", "color(s)"/"animated shapes"/"abstract" topics, and media such as "film". Yet, their own results point to broader imaginative elements that extend beyond VMI alone. Other emergent thought types—fictional imaginings, autobiographical recall, affective and sensory experiences—can be seen to form integral parts of listeners' reported experience. Hashim et al. found participant descriptions often included non-visual elements like affect, semantic, and musical detail despite being instructed to only focus on visual imagery. They themselves note "narrowing one's research aims on just the visual components of a listeners' experience [...] overshadows its potential unique links to other semantic associations formed in response to the music" (Hashim et al., 2023, p.15–16)—multi-modality continues to be found to be an important and inherent part of the VMI experience (Cespedes-Guevara & Dibben, 2022; Deroy, 2020; Nanay, 2018)

Furthermore, existing work like the studies reviewed above (Dahl et al., 2023; Küssner & Eerola, 2019; Hashim et al., 2023) have generally used limited ranges of musical stimuli, both in clip numbers (e.g. using three, four, or even zero clips (reliance on recall)) and stylistic variation (e.g. only film music). However, musical diversity is crucial for capturing a fuller spectrum of MIMC, which may differ significantly between genres and personal preferences of listeners. Genre has been shown to significantly influence the occurrence and type of thoughts (van der Walle et al., 2025b). Moreover, variations in listeners' contextual associations with music—from cultural and historical, to personal and prior experiences—likely play an important but underexplored role in shaping the qualities and content of our thoughts (Thompson et al., 2023; Turnbull, 2025).

Present study

Building on these foundations, we applied manual qualitative analysis to free-response reports of MIMC across a wider range of musical styles to address key gaps left by more constrained, rating-based, or computational approaches. Rather than focusing on a single, predefined thought type, this study examines all MIMCs reported by listeners. To support this breadth, we used a large and stylistically diverse set of musical stimuli: 120 clips spanning 10 genres. Such diversity offers a naturalistic way to cover a wide spectrum of acoustic properties, emotional expressions, and contextual associations. We recruited participants balanced across adulthood, aged 18–75 years, allowing us to consider variations spanning multiple generations and simulating the diversity of music that people may encounter in everyday life.

We analysed MIMC descriptions using an inductive approach, given the lack of an overarching theory on MIMC. Qualitative coding methods including thematic and content analyses were employed to identify categories, patterns, and themes within and across musical styles and participant groups. This enabled us to capture subtleties in MIMC descriptions that might be obscured by purely quantitative or automated methods. Our aim is to provide a more comprehensive and nuanced picture of listeners' "thoughtscapes" (van der Walle et al., 2025b), highlighting the richness and diversity of people's imaginative experiences during music listening. We wish to lay the groundwork for future models of MIMC that are more contextually sensitive, considering a fuller spectrum of people's imaginative experiences during music listening.

Method

Materials/Stimuli

Music stimuli

Musical stimuli were originally taken from the MUSIFEAST-17 dataset (van der Walle et al., 2025a), which comprises 30-s instrumental sections from commercially released music. We chose 10 of the 17 genres encompassing a wide range of styles—Ambient, Classical, Electronic, Film, Funk, Jazz, Metal, Pop, Sixties pop, and Eighties pop—using MUSIFEAST-17's accompanying normative data to inform our stimuli choices as follows. Some clips were amplitude-normalised prior to use. As a result, said clips are not identical to the original MUSIFEAST-17 clips, having altered acoustic properties relative to the original dataset. In line with open research practices, explicit documentation, including both original references and modified versions, are publicly available in our repository https://osf.io/aepy3.

We ensured the genres chosen had a moderate rate of different thought types (in response to MUSIFEAST-17's question "*Which of the following categories of thoughts did you have while listening to the music? Feel free to choose more than one option if you had multiple types of thoughts*"), excluding music clips with high rates (>20%) of proxied mind-wandering thoughts ("*I was thinking about everyday stuff*", "*I had thoughts about the future or personal plans*", "*no thoughts*") to decrease the chance of such thoughts in the present study. Clips often rated (>50%) as not originating from Europe or USA/Canada (in response to MUSIFEAST-17's question "*Where do you think this music was created?*") were excluded as the present study aimed to study Western (UK and US) listeners' thoughts during music representative of their typical everyday listening context. We also included popular music from different time periods—Sixties, Eighties, and modern Pop—to explore whether this temporal distinction impacted MIMC. To ensure the three pop subgenres were differentiated and aligned with the intended time period, we excluded modern Pop clips released before the year 2000 and used MUSIFEAST-17's decade-identification rates to exclude clips with low correct decade-identification rates (<12.5%).

For each of the 10 genres, 12 clips were selected, comprising the clips with the 6 highest and 6 lowest mean familiarity ratings from the remaining pool in the MUSIFEAST-17 dataset following the above exclusion steps. These 120 clips were divided into 12 stimulus groups, each containing 1 clip from each genre (10 clips per group) and an equal split of high and low familiarity clips. Participants were randomly assigned to one of these stimulus groups upon entry in the study, meaning they each heard one clip from each genre.

Main study questions and tasks

The study was created and run online using *PsychoPy* and *Pavlovia*. After hearing each music clip, participants were first asked whether or not they imagined any thoughts or scenes while listening to the music. If they responded "yes", they were presented with an open-response question asking: "Please describe the thoughts/scenes you imagined in as much detail as possible, but do not spend more than about 1 minute on the response." Responding "no" also prompted an open-response question, asking them to explain why they might not have imagined any thoughts or scenes (this was to prevent participants from responding "no" to advance through the experiment more quickly). Participants also rated each thought description and music clip on several Likert scales (e.g. spontaneity of thoughts, familiarity of music). In the present paper, we focus on the contents of participants' open-ended MIMC descriptions—other aspects are analysed further in separate publications. At the end of the study, participants completed final demographics questions. All survey questions are included as Supplemental material in the repository.

Participants

Participants were recruited, screened, and compensated (£3.75) through *Prolific*. Stratified sampling was used to recruit an even balance of participants across four age groups (18-30, 31-45, 46-60, 61-75 years), female/male sex representation[1], and UK and US nationality.

In total, 1,872 participants completed the study. We excluded 82 participants who failed the attention check (see Procedure), and removed duplicate data from two participants, retaining only the earlier timestamped completion for each Prolific ID. Two further participants were removed—one for mentioning use of an AI assistant and one for providing extensive task-irrelevant commentary about current events in their open-text responses—resulting in 1,786 retained participants. Within these retained cases there were 19 individual missing open-text responses (entries of "NA" variants or unfinished sentences); because no participant had more than one such missing trial, we excluded only the singular missing data trials rather than entire participants, yielding 17,841 complete trials for analysis.

The 1,786 retained participants ranged in age from 18 to 75 years (M = 44.39, SD = 15.76; 887 female, 884 male, 11 other, 4 prefer not to say) with a comparable split between the four age groups (18–30 N = 471, 31–45 N = 458, 46–60 N = 440, 61–75 N = 417). All participants reported current residency in the UK (N = 910) or US (N = 876) and 1731 (96.9%) reported English as their native language. 24.2% of participants held qualifications equivalent to UK A-Level, Scottish Highers, or US High School Diploma, 40.3% had completed an undergraduate degree, and 16.6% had completed graduate degrees. All but 7 participants reported wearing headphones for the full study duration (99.6%). A small proportion reported some level of hearing impairment (4%); none of these participants were excluded as hearing impairments cases were mild to low level or used corrective measures[2], they all passed the initial headphone check, the auditory attention check, and didn't report any difficulty hearing the music. Using the singular musicianship identification self-report item from Ollen (2006; Zhang & Schubert, 2019), 16.5% classified themselves as non-musicians, 58.4% as music-loving nonmusicians, 16.2% as amateur musicians,

[1] All gender representations were accepted, including "Other" and "Prefer not to say" options in the demographics questions.

[2] Examples of hearing impairments and corrective measure descriptions included "Tinnitus, no measures to correct it", "i have some earwax blockage in my left ear.", "Mild age related hearing loss. Had no problems hearing the music.", "Otosclerosis. I wear hearing aids. My phone and this laptop are connected to them via bluetooth.", and "Age related high frequency loss. No aids-just turn the volume up!"

4.9% as serious amateur musicians, 2.6% as semi-professional musicians, and 1.4% as professional musicians.

Procedure

Upon entry, participants were randomly assigned one of the 12 stimulus groups using *Pavlovia*'s online counterbalancing feature so each clip would be heard by a balanced number of UK and US participants across the four age groups. Each stimulus group was heard by 140 to 153 participants in total. Before starting the experiment, participants were informed that keyboard responses were required for the study so could only be completed on a computer or laptop (not a mobile device or tablet). Instructions to wear headphones followed with a sound check by listening to two 7-s clips, one quiet and one loud relative to the stimulus set, to prepare participants' device volume for the study. They were instructed to set their device to a volume at which both music clips were comfortably heard and to maintain this volume for study duration. To ensure participants were wearing headphones and that they had sufficient sound quality, a technical headphone screening task (Woods et al., 2017) had to be completed.

To begin the main experiment, the study information sheet was presented and informed consent was required to continue. Instructions and guidance was given about the main study, asking participants to relax, listen without distraction in a quiet environment, that there were no "right" answers, and having no thoughts was valid and should be reported. For each experimental trial, participants listened to a 30-s music clip from their assigned stimulus group (presented in a randomised order) and completed the clip response questions after each clip (see Supplemental material). After the fifth trial, an attention check was used to ensure participants were paying attention and hearing sound through their device. A computer-generated spoken word track asked "Please type the word banana in the textbox below", presented only with a textbox on a blank screen, progressing automatically after 20 seconds. Upon completing each of the 10 experimental trials, participants completed the demographic questions.

The time limit for open-text thought descriptions was set with a minimum of 30-s and maximum of 2-minutes. This constraint helped standardise response length while encouraging thoughtful engagement without excessively prolonging the task. Participants were informed of the time limits. Apart from the 30-s clip listening durations and 20-s attention check cut-off, no other time limits were set for the study allowing participants to go through the questions at their own pace. Regarding online participation controls, *Prolific* has measures in place to stop participants from leaving and returning to a study, and for a study estimated to take 30-minutes, *Prolific* set a maximum completion time of 87-minutes.

Analysis

Mixed-methods analyses were employed. Primarily, qualitative analysis was used to investigate the contents, themes, and patterns of participants' open-text MIMC descriptions. Therefore, the main analysis focuses on the trials where participants had responded "yes" to having imagined any thoughts or scenes, but all trials were coded and considered in the final coding pass and discussion. Analyses were initially conducted on *NVivo* (15.2.0, 15.3.0), using *MAXQDA* (26.1.0) for the final coding pass. To complement the qualitative analyses, final code allocations were quantised to examine structural relationships between categories using co-occurrence and network analyses. Quantitative analyses and visualisations were conducted on *RStudio* (2026.01.1+403). All datasets and materials generated in this study are available in our repository (https://osf.io/aepy3).

Qualitative

An inductive, mixed-methods approach was used in our qualitative analyses, deriving the codes from the raw data, prioritising free emergence of the final narrative without forcing a theory on the data (Glaser, 1978, as cited in Qureshi & Ünlü, 2020). An iterative approach to coding and categorisation was applied throughout as the coders learnt more about the qualitative dataset holistically while working through the coding and categorisation stages (Locke et al., 2022). Reflexivity was required of all coders, being aware of their own biases and interpretive notions, encouraging reflective memos to clarify meanings and reasoning behind code decisions (Bryman, 2016; Glaser & Strauss, 2017). To enhance trustworthiness we maintained an auditable decision trail (codebook versions, joint *NVivo* files, meeting notes), following and demonstrating recommended procedures as in Nowell et al. (2017). Our analysis procedure is broken down into three stages below.

Stage 1. The lead investigator conducted an initial coding pass on a random 20% sample of participants balanced across age, nationality, and stimulus group. Initial codes were generated *in vivo*—using participants' own words to capture MIMC descriptions as concise codes (Bazeley & Richards, 2000). Simultaneous coding was permitted where a single thought description warranted multiple codes. These in vivo codes were then organised into categories and subcodes, grouping items by similarity or conceptual relation to form an initial codebook (Kuckartz & Rädiker, 2023). A second coding pass on the same 20% sample served to refine the codebook; codes were iterated, rearranged, and clearly defined to ensure hierarchical fit and operational clarity.

Stage 2. Five research assistants each coded separate 20% portions of the dataset alongside the lead investigator to enable team-based checks, improve analytic dependability, and further refine the codebook (Elo et al., 2014; Guest et al., 2012). All coders used the refined codebook in completing the third coding pass; *Nvivo*'s *Coding comparison query* feature was used to quantify agreement. Codes with percentage agreement below 90% were discussed in one-on-one meetings between each research assistant and the lead investigator (McHugh, 2012); consensus was reached through two-way negotiation where both parties revised codes and collaboratively improved categories. The lead investigator reviewed all one-on-one meeting notes and revised the codebook accordingly; the revised codebook was then trial-coded and iteratively tested following Schreier's (2012) recommendations to refine and strengthen the final codebook. An auditable decision trail, including intercoder agreement and codebook iterations, is openly accessible in our repository.

Stage 3. The lead investigator applied the final codebook across the full dataset in a final coding pass to conduct qualitative content analysis (conventional, inductive) and reflexive thematic analysis in parallel to identify recurring patterns, concepts, and overarching themes across the dataset. Content analysis procedures (Hsieh & Shannon, 2005; Schreir, 2012) provided a systematic account of category frequencies and supported transparent mapping from in vivo codes to higher-order categories. Reflexive thematic analysis followed Braun and Clarke's phase-based model—familiarisation, initial coding, searching for candidate themes, reviewing and refining themes, defining and naming themes, and producing the analytic narrative—emphasising the researchers' interpretive role and iterative construction of themes (Braun & Clarke, 2006; Braun & Clarke, 2021). In this framework, categories functioned as nested sub-codes and themes represented higher-order conceptual groupings across the dataset. The combined content-analytic and reflexive thematic approach allowed us to report both systematic category distributions and richer interpretive themes grounded in participants' language.

Quantitative

Complementary analyses involved transforming the qualitative coding into a quantised dataframe; each MIMC was represented as a binary vector indicating the presence or absence of every code level (themes, categories). This quantisation enabled systematic computation of code co-occurrence, both in relative terms (Jaccard similarity) and absolute terms (raw co-occurrence proportions). Jaccard was preferred as the primary metric because category base rates of occurrence varied considerably, meaning raw binary proportions inflate apparent co-occurrence for frequent categories regardless of genuine associative strength; Jaccard corrects for this by expressing joint occurrence relative to union occurrence. Raw proportions are nonetheless reported in parallel to allow direct comparison of absolute and relative overlap. These metrics were used to construct a co-occurrence matrix, allowing us to identify which categories most frequently appeared together and which operated more independently.

Network analysis techniques were applied to the co-occurrence matrix to explore the relational architecture of the coding frame. Using force-directed (Fruchterman–Reingold; Fruchterman & Reingold, 1991) layout with Jaccard distances, we generated a co-occurrence network: edge thickness reflected similarity (based on Jaccard co-occurrence), node size reflected prevalence, and node position reflected the force-directed embedding based on Jaccard similarity (more similar nodes positioned closer together). This approach enabled observation of higher-order patterns not easily visible through qualitative reading alone such as bridging roles and relative peripherality of categories. These quantitative patterns informed the interpretive analyses reported in the Results, providing a mixed-methods account of how different forms of music-influenced mental content tend to occur and organise into broader thematic structures.

Results

Experiencing music-influenced mental contents

Across all valid trials (N = 17,841), 14,091 MIMCs were reported and described, giving a reporting frequency of 78.98% overall. Out of 10 possible trials, participants reported M = 7.89 (SD = 1.77) thoughts, with thought description word counts averaging M = 17.18 (SD = 10.58, *range* 1–111). Thought reporting rates between genres differed significantly but with a small effect size ($\chi^2(9, N = 17841) = 309.06$, $p < .001$, $V = 0.132$); Film (86.8%) and Jazz (86.1%) had the highest rate of reported thoughts, and Electronic (71.5%) and Pop (70.9%) had the lowest. Reporting rates were comparable across geographical groups, with UK listeners reporting thoughts on 78.4% of trials and US listeners on 79.6%. Similar consistency was observed across age groups; 18–30 (78.9%), 31–45 (77.7%), 46–60 (79.2%), and 61–75 (80.3%).

Thematic analysis of music-influenced mental contents

From the coding passes and codebook refinement, nine final categories were established. These nine categories fall into three top level themes: *Narrational*, *Referential*, and *Experiential*. Each theme is broken down into its respective top-level categories in the following subsections, providing short definitions and three exemplar quotes (anonymised: *Participant ID—genre*). Co-occurrence analyses (Figure 1 and 2) of top-level categories and their parent themes (prioritising Jaccard similarity values) inform the observed patterns and are interpreted within each section, highlighting the relational network between different forms of MIMC.

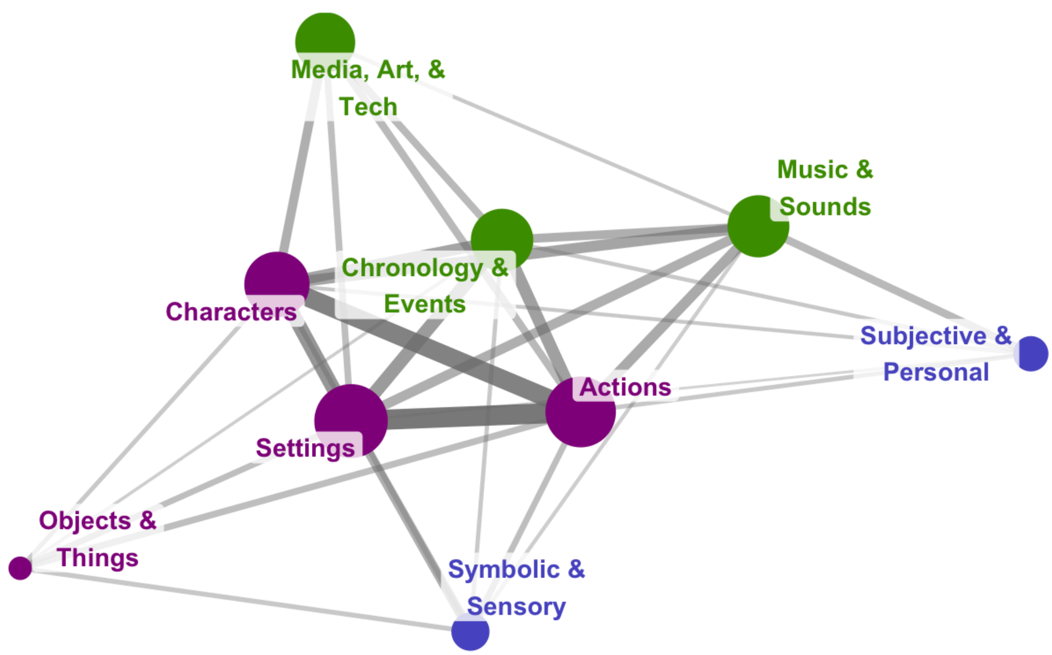

Fig. 1 Co-occurrence network analysis of MIMC categories. Fruchterman-Reingold layout. Nodes that cluster spatially co-occur frequently using Jaccard distance matrix (positions reflect pairwise similarity). Edge threshold ≥0.11. Node colour = theme (purple = *Narrational*, green = *Referential*, blue = *Experiential*), node size = prevalence, edge thickness = Jaccard similarity (strength of co-occurence).

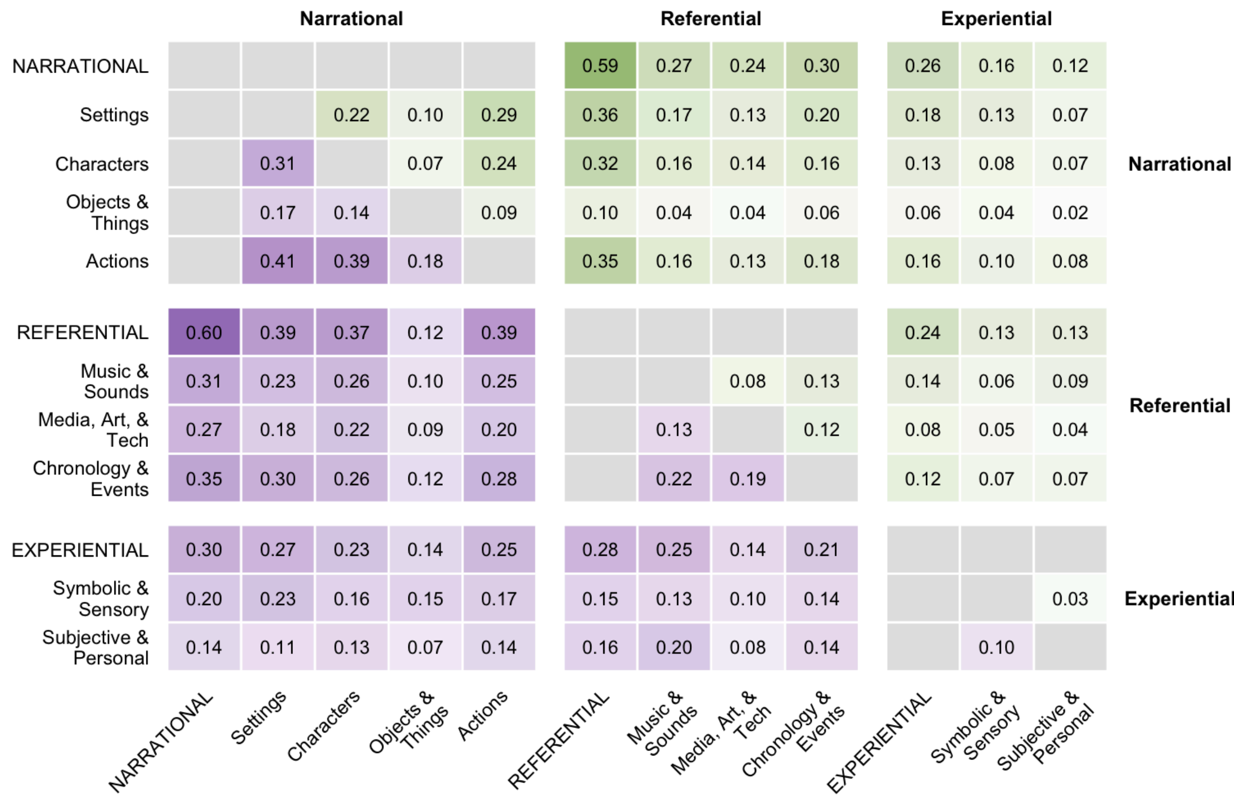

Fig. 2 Co-occurrence matrix of MIMC themes and categories. Upper green triangle = raw co-occurrence proportion, Lower purple triangle = Jaccard similarity, Grey tiles = diagonal or parent-child comparisons (same-same not informative).

Narrational

The *Narrational* theme captures concrete, scene-like elements in participants' MIMC descriptions—the *where*, *who*, *what*, and *how* aspects that together form short stories. These elements were common, appearing in 79.8% of MIMC descriptions across the dataset, and often provided the scaffolding on which more *Referential* or *Experiential* content was layered. Quantitatively, *Narrational* categories formed a densely interconnected cluster in the co-occurrence networks: *Settings*, *Characters*, *Objects & Things*, and *Actions* showed the strongest within-theme co-occurrence of any part of the coding frame (e.g., *Settings* × *Actions* J = .41; *Characters* × *Actions* J = .39; *Settings* × *Characters* J = .31). In the network layout, these four categories sit in close spatial proximity, forming a *Narrational* core that anchors the broader MIMC landscape. Table 1 provides a compact overview of *Narrational*'s four top-level categories.

Table 1. Narrational theme's four top-level categories. Second-order subcategories named with accompanying example content. Dataset coverage percentage relates to the overall top-level category.

Top-level category	*Subcategory*	*Example content*	*% dataset coverage*
Settings	Natural world	terrain, water-features, weather, sky, space, creatures (real or mythical)	19.4%
	Fictional and Spiritual places	post-apocalyptic, enchanted, virtual, Heaven	1.4%
	Societal places	café, theatre, kitchen, temple	29.6%
	Named regional place	countries, cities, landmarks	5.6%
	Scene qualities	gloomy, bright, pastoral	9.1%
	Spatial orientation	up, around, under, nearby	4.0%
			∑ = 51.3%
Characters	Character descriptors	outfits, identity, archetypes, traits	35.4%
	Named figures	real-world, historical, religious, fictional	4.9%
	Social presence	groups, crowds, couples, alone	6.2%
	Body parts	brains, fingers, ankles	2.3%
			∑ = 40.3%
Objects & Things	Consumables	food, drink, drugs	3.3%
	Everyday objects	public and domestic items, furnishings, props	5%
	Transport	real or fictional vehicles, features, modifications	5.8%
	Commerce and Organisations	logos, corporations, institutional entities	0.8%
	Materials and Surfaces	silk, cobblestone, mirrored	0.9%
	Object qualities	size, condition, style	0.7%

			∑ = 14.6%
Actions	Agentic actions	purposive, embodied, socially enacted, *describe what is being done*	35.2%
	Processes and States	attentional, temporal, processual states, *describe what is happening*	17.6%
	Action modifiers	characterisation, speed	4.3%
			∑ = 47.1%

Settings

Settings captures landscapes, locations, and atmospheres—real, imagined, natural, societal, and stylistic. Concreteness varied wherein some participants described precise locales and others offered impressionistic atmospheres nonetheless anchoring the MIMC.

Natural world elements included real and fictional *creatures* ("bulldog", "dragon"), *terrains* ("tundra"), *vegetation* ("willow"), *water features* ("rapids"), *sky and space* ("clouds", "cosmos"), and *weather* ("rainfall"). *Fictional and Spiritual places* encompassed wonderous, magical, futuristic, and religious places ("enchanted forest", "virtual worlds", "hell", "other-worldly"). *Societal places* covered human-made infrastructure ("home", "pub", "highway", "airport"). *Named regional places* ranged in geographic scale ("Latin America", "Morocco", "Glasgow") and locational specificity ("Monument Valley", "White Cliffs of Dover", "Stubb's Bar-B-Q"). *Scene qualities* denoted the scene's stylistic tone ("serene", "mysterious", "trippy") and perceived scale or density ("vast", "crowded"), while *Spatial positions* provided placement or perspective ("beside", "around", "distant").

Scene setting often functioned as the stage and structural base layer for other *Narrational* elements, reflected in its high co-occurrence with *Actions* (J = .41) and *Characters* (J = .31). In the network diagram, *Settings* appears as one of the largest and most central *Narrational* nodes, acting as a bridge node to *Referential* (J = .39) and *Experiential* (J = .27) themes. *Settings* also exhibits strong links to *Chronology* (J = .30), *Symbolic* (J = .23), and *Music* (J = .23), indicating that participants frequently grounded their descriptions in a place while simultaneously describing temporal, abstract, or musical qualities.

Exemplar quotes

- *P1018—Film:* "Pushing through the leaves and stumbling into a massive, green glade. The splendor of the natural world on full display everywhere you look. Deer and other animals drink from a crystal clear lake, squirrels scurry up trees, and so on."
- *P278—Ambient:* "I imagined floating or flying silently through space. Planets and stars passing by slowly"
- *P487—Jazz:* "[...] scenes of European destinations - like roads and towns and coastal scenes of places in France or Italy, with sun shining and seas blue."

Characters

Characters captures real and fictional people, identity, appearance, roles, and social configurations. *Character descriptors* included *outfit details* ("flares", "sunglasses", "afro"), *identity* ("Arabic", "Gen-X", "yuppies"), *professions and pursuits* ("bartender", "drummer", "producer"), *relations* ("mom", "mates", "neighbour"), *archetypes* ("antagonist", "killer", "final boss"), and *traits* ("wise", "drug addled"). *Named figures* included *fictional characters* ("Batman", "Big Bird", "Romeo"), *historical and mythical figures* ("Napoleon", "Ba'al"), *performers and entertainers* ("Uma Thurman", "Brock Lesnar"), and *creators* ("Hitchcock", "Miyazaki"). Explicit indications of *Social presence* configurations ranged from *groups* ("with friends", "with a ton of strangers"), to *dyads* ("two siblings", "with the woman I love"), to *solitude* ("solo", "nobody"). *Body parts* captured literal anatomical references ("fist", "fingers", "flesh"), often appearing in disembodied or surreal MIMCs.

Characters frequently carried affective cues and action modifiers (e.g., "determined", "forlorn"), linking them tightly to *Actions*, reflected in their co-occurrence (J = .39) and adjacency in the network layout. *Characters* sits between *Narrational* and *Referential* clusters, with a strong cross-theme co-occurrence (J = .37). Named figures often indicated cultural referents or generic media tropes, consistent with *Characters*'s links to *Chronology* (J = .26), *Music* (J = .26) and *Media* (J = .22).

Exemplar quotes

- *P274—80s-pop:* "Miami-set 80s cop thriller. Men in shades and white baggy trouser/jacket combos tracking along the edge of a building or sidewalk with their guns drawn. Elvis the Alligator an optional presence."
- *P1597—Metal:* "My head bursting open and pieces of my brain are flying out into the air around me making a circle going faster and faster around my head."
- *P1420—Film:* "[...] Theodore Twombley is alone with his thoughts and no one to talk to. He has to navigate these feelings of lonliness and freedom to see the good and bad in his life."

Objects & Things

Objects & Things contains non-living entities, items, and materials. Objects supported imagined settings in tangible detail, functioning as scene anchors or symbolic meaning carriers by cueing inferred histories, social status, or (un)reality.

Subcategories included *Consumables* ("coconut", "whisky", "ecstasy"), domestic and public *Everyday objects* ("sofa", "statue", "sword"), *Transport*—from ordinary to cinematic or fantastical ("truck", "hydraulics", "UFO"), and *Commerce and Organisations* ("icons", "phone line", "industry"). *Materials and surfaces* ("voile", "sandstone") and *Object qualities* ("massive", "spiky", "whimsical") further texture these artefacts, signalling era, subculture, and modulate intensity or density. *Objects*, while rare, were most commonly described within settings to create atmospheric tone and shape how expansive or contained a scene felt.

Objects showed moderate co-occurrence with *Settings* (J = .17) and *Actions* (J = .18), and sits slightly peripheral to the core *Narrational* triad (*Settings–Characters–Actions*) in the network diagram, reflecting their role as supporting scene elements rather than primary narrative drivers. *Consumables* and generic objects appeared most often in *Experiential* MIMC (supported by co-occurrence J = .14), especially when describing surreal visuals or introducing sensory immediacy, consistent with their relatively moderate co-occurrence with *Symbolic* (J = .15). Objects denoting public or domestic spaces situated MIMC in intimate, habitual, or leisure spaces, while *commerce* shifted the frame toward economic or cultural domains, reflected in its co-occurrence with *Settings* (J = .17). *Transport* frequently implied motion and/or story progression, co-occurring with *Actions* (J = .18) and *Chronology* (J = .12), forming travel and escape micro-narratives.

Exemplar quotes

- *P342—Classical*: "[...] in a shakespeare play or attending a tudor medieval party, drinking ale from a metal mug and eat[ing] soup with bread."
- *P1579—Ambient*: "[...] on a phone waiting for customer service [...]"
- *P80—80s-pop*: "[...] you proceed to head into your Lamborghini Countach/Ferrari 250 GTO and drive off [...]"

Actions

Actions captures verbs and modifiers that animate scenes for both animate and inanimate elements. It comprises three third-level categories: *Agentic actions* describing purposive, goal-directed behaviours, often propelling narrative development; *Processes and States* capturing attentional, resting, processual, and temporal states; and *Action modifiers* qualifying and recalibrating scene animation.

Agentic actions included *directed movement* ("leaving", "driving"), *daily life* ("errands", "hustling"), *music-linked movement* ("moshing", "tapping"), and *sports* ("parkour", "rafting"), to *creating and performing* ("jamming", "choreographing"), *relational and social* ("teach", "flirting"), *consuming* ("sipping", "smoking"), and *conflict and wrongdoing* ("abuse", "fight"). *Processes and States* encompassed *perceptual acts* ("gazing", "pondering"), *non-directional motions* ("floating", "dripping"), *rest and leisure* ("gaming", "sleeping"), *transforming and learning* ("discovering", "exploding"), and *waiting and anticipating* ("preparing", "expecting"). *Action modifiers* described *speed* ("instantly", "sluggishly") and *characterisation* ("flamboyantly", "aggressively").

Actions is the most interconnected *Narrational* category, with strongest within-theme links to *Settings* (J = .41), *Characters* (J = .39), and *Objects* (J = .18), supporting its role as scene animator of living and non-living elements. In the network layout, *Actions* occupies a centre position within the *Narrational* cluster, reflecting its role in driving narrative progression. Directed movement often produced high-agency narrative peaks, whereas processual motions marked pauses, transitions, or internal shifts in narrative flow—jointly corroborated by *Action*'s strong link with *Chronology* (J = .28). *Music-linked movement* appeared both as third-person animation and embodied expressions of urges to move ("made me want to dance"), frequently paired with positive *Opinions and Preferences* (see *Subjective & Personal experience*) of the music. Both agentic and processual actions often mirrored music's *structural and technical notes* (e.g., tempo, beat; see *Music & Sounds*) ("walking briskly travelling just as quick as the progression of the music". Overall, *Actions* functions a hub node, linking *Narrational* content with *Referential* (J = .39) (*Music & Sounds*; *Media, Art, & Tech*) and *Experiential* (J = .25) (*Symbolic & Sensory; Subjective & Personal experience*) themes, reflecting how action descriptions often blend narrative, musical, and affective dimensions of MIMC.

Exemplar quotes

- *P455—Funk*: "[...] a man wearing a red silk gown, schmoozing around trying to seduce a lady [...]"
- *P1599—Pop*: "[...] everyone is just dancing [...] probably under the influence of drugs like ecstasy and alcohol"
- *P1578—Film*: "[...] working out in my home office - just that feeling of working my butt off to conquer the world while living and working in the NYC metro area when I was fresh out of college."

Referential

The second theme, *Referential*, captures aspects of participants' MIMC descriptions that alluded to, drew on, or explicitly named an extramusical source. In simpler terms, these are descriptions where participants used references—genres, media, cultural artefacts, historical periods, or seasonal markers—to articulate what they were experiencing or thinking. *Referential* content appeared in 77.9% of MIMC descriptions and is organised into three top-level categories.

Co-occurrence patterns show that *Referential* categories form a cohesive interpretive triad, with internal connections between *Chronology* and *Music* (J = .22), *Chronology* and *Media* (J = .19), and *Music* and *Media* (J = .13). Across themes, Referential content showed strongest bridging to *Narrational* content (J = .60), especially *Settings* (J = .39), *Actions* (J = .39), and *Characters* (J = .37). In the network layout, the three *Referential* nodes cluster together but sit adjacent to the *Narrational* core, reflecting their role as interpretive overlays that contextualise and frame the unfolding scene.

Table 2. Referential theme's three top-level categories. Second-order subcategories named with accompanying example content. Dataset coverage percentage relates to the overall top-level category.

Top-level category	*Subcategory*	*Example content*	*% dataset coverage*
Music & Sounds	Instruments, Ensembles, and Imitations	saxophone, a cappella, orchestra, baton, pretending to play drums	8.8%
	Music use and function	song for chores, sell a product, gets you pumped, for the soul	2.4%
	Sound qualities	how music sounds/feels; genre/cultural/stylistic label; structural and technical notes	9.0%
	Playback and Listening devices	cassette, CD, boombox, car radio, headphones	1.6%
	Named Artists and Musicians	Nina Simone, Prince, Avicii, Lady Gaga, the Rat Pack	5.8%
	Named Songs and Lyrics	Enter Sandman, The Girl From Ipanema, "sweet dreams are made of this"	2.1%
	Sounds and Sound effects	barking, beeping, revving, boom, noises	12.9%
			∑ = 36.4%
Media, Art, & Technology	Media formats	musical, talk show, advert, VHS, book	26.0%
	Media genres and styles	Bollywood, thriller, RPG, cheesy, feel-good	8.0%
	Media packaging structures	opening credits, cliffhanger, cutscenes	3.5%
	Media techniques and effects	close-up, panning, green screen, slowmo	2.4%
	Music and sound in media	soundtrack, theme tune, voiceover	4.9%
	Media companies and Devices	named corporations, publishers, streaming services, hardware brands	1.3%
	Digital platforms and	online forum, social media, apps, AI	0.8%

	Applications		
	Named media references	Sesame Street, BBC-Proms, Minecraft, "a long time ago in a galaxy far, far away..."	11.2%
			∑ = 34.3%
Chronology & Events	Events and Occasions	quinceañera, funeral, festival, sports match, therapy session, exam	12.1%
	Historical events and periods	Early-2000s, Swinging Sixties, wartime, prohibition, ancient times	12.9%
	Narrative progression	impending doom, pivotal scenes, climax	5.6%
	Passage and Repetition of Time	all night, passing time, every single time	3.6%
	Seasons and Time	morning, 3am, Autumn, weekend	9.2%
			∑ = 37.0%

Music & Sounds

Music & Sounds contains references to music, sound, and auditory experience. These span instruments, creators, playback, qualities, functions, and non-musical sound events.

Relating to the sound itself, references encompassed *Instruments, Ensembles, and Imitations* ("duet", "brushes", "air guitar"), *Sound qualities* ("spooky", "EDM", "beat"), *Sounds and Sound effects* ("chirping", "sirens", "commotion"), and specific *Named Artists and Musicians* ("SOAD", "Debussy", "Gaga") and *Named Songs and Lyrics* ("Take On Me", "Fly me to the moon"). Referenced extramusical aspects included *Music use and function* ("maintain a buzz", "keep me concentrated"), and *Playback and Listening devices* ("records", "stereo", "earphone").

Music & Sounds was a well interconnected *Referential* category, with strong cross-theme co-occurrence with *Narrational* content, particularly *Settings* (J = .23), *Characters* (J = .26), and *Actions* (J = .25), with a smaller but notable link to *Actions* (J = .10). This pattern reflects how participants often described musical qualities alongside vivid scene elements, tightly binding sound to narrative content. Internally, *Music* was most strongly linked to *Chronology* (J = .22) and

Media (J = .19), suggesting that perceived narrative progressions were often shaped by musical features and framed through media or genre templates. Its moderate to *Experiential* categories (*Symbolic* J = .13; *Subjective* J = .20) indicate that sound-based references sometimes carried sensory metaphors and meaningful, personal connections. In the network layout, *Music & Sounds* is positioned slightly on the periphery but acting as a key bridge node to all three themes.

Exemplar quotes

- *P1123—Jazz*: "Moody jazz club lighting, I could visualize the drummer on the cymbals tss-ts-ta-tss-ta-ta'ing. Fedoras and blazers. Candles on tables."
- *P977—Metal*: "EXIT LIGHT[ ]ENTER NIGHT[ ]TAAAAAAAAAAKE MY HAND[ ]WE'RE OFF TO NEVER[-]NEVER[ ]LAND"
- *P1741—60s-pop*: "[...] I have a bit of a distance to go but it's all good because the music makes the distance tolerable. I am walking to the beat of the music."

Media, Art, & Technology

Media, Art, & Technology includes references to any type of media, artistic, entertainment, and technological domains. These include media formats, platforms, named works, techniques and effects, present in digital or analogue forms.

Subcategories included generic descriptions of *Media formats* ("sitcom", "comic"), *Media genres and styles* ("sci-fi", "gritty"), *Media packaging structures* ("titles", "montage"), *Media techniques and effects* ("wide shot", "CGI"), and *Music and sound in media* ("OST", "narration"). More specific references included *Media companies and Devices* ("Atari", "SNES", "Spotify", "VR headset"), *Digital platforms and Applications* ("Mindful", "meme", "YouTube"), and *Named media references* ("Blade Runner", "Dungeons & Dragons", "Gotham").

Media, Art, & Tech showed moderate internal connectivity within the *Referential* theme, with its strongest link to *Chronology* (J = .19) and a smaller connection to *Music* (J = .13). Cross-theme, it co-occurred most strongly with *Narrational* categories, particularly *Characters* (J = .22), *Actions* (J = .20), and *Settings* (J = .18), indicating that media references often shaped or were used to describe how scenes and characters were experienced and animated. Qualitative inspection of coded excerpts suggests that such references often provided stylistic framing for MIMC rather than functioning as standalone content (see exemplar quotes below). Its links with

Actions (J = .20) and *Chronology* (J = .22) further reflect participants' use of media tropes to describe movement and pacing, such as cinematic transitions and game-like sequences. Weaker connections to *Experiential* categories (*Symbolic* J = .13; *Subjective* J = .08) suggest that media references occasionally carried symbolic or personal meaning but, compared to its function as interpretive or stylistic colouring within *Narrational* and *Referential* MIMC, was less relied upon in *Experiential* MIMC.

Exemplar quotes

- *P851—Metal*: "I envisaged this as part of a movie, a dark cityscape, lots of lights and neon, person standing by themselves looking around then the camera draws back and there is a car racing towards the person, a fast sports car, sliding on the wet road, going faster and faster aiming to put our hero in danger."
- *P612—Electronic*: "Alien drones and acidic bass for a dark opening, science fiction opening scenes before the chase begins"
- *P1317—Pop*: "I imagined a car commercial, with a guy speeding down the road. I also imagined a drink commercial for alcoholic beverages with a guy shown in a bar talking to a girl and looking confident. The music exuded confidence and masculine energy to me. "

Chronology & Events

Chronology & Events contains references to time, temporality, and happenings—from daily markers to major historical events and narrative structures. Subcategories comprise *Events and Occasions* ("yoga class", "parades", "shows"), *Historical events and periods* ("golden age", "Renaissance", "Vietnam War"), *Narrative progression* ("build up", "triumph", "tragic moment"), *Passage and Repetition of Time* ("spending time", "everyday", "million times"), and *Seasons and Time* ("midnight", "Winter", "Christmas").

Chronology & Events showed the strongest internal connectivity within the *Referential* theme, especially with *Music & Sounds* (J = .22) and *Media, Art, & Tech* (J = .19). Cross-theme, it strongly co-occurred with *Narrational* (J = .35) categories including *Settings* (J = .30), *Actions* (J = .28), and *Characters* (J = .26), indicating that participants often embedded their scenes within a temporal frame or imagined events unfolding over time. Its links to *Experiential* (J = .30) categories, *Symbolic* (J = .20) and *Subjective* (J = .14), suggest that temporal framing could carry symbolic or emotional meaning, such as seasonal moods or life-stage reflections. In the network diagram, *Chronology & Events* sits centrally within the *Referential* cluster, organising MIMCs into narrative arcs that connect sound and media to scenes.

Exemplar quotes

- *P1018—Ambient*: "The first sunrise of all time, shedding light and warmth upon denizens of a world that had previously only known darkness. The coming of a new era, where things will never be as they were ever again."
- *P1289—Electronic*: "Cold winter lands. Dawn of the space age. Technological advancement. A satellite in low earth orbit."
- *P176—60s-pop*: "I imagined the leader of a gang finally getting his break, only after shooting his rivals all alone and walking his way out with the sun hitting him and then smiling to himself as he's finally done with this business."

Experiential

Lastly, the theme *Experiential* captures subjective and figurative elements within participants' MIMC descriptions, including ideological values, sensory oddities, abstractions, opinions, reactions, and personally meaningful connections. This theme was the rarest and most intimate recountings of participants' MIMCs, appearing in 31.2% of descriptions, reflecting participants' attempts to articulate how the music felt, what it meant, or how it resonated with them. *Experiential* content is expressed through two top-level categories.

Table 3. Experiential theme's two top-level categories. Second-order subcategories named with accompanying example content. Dataset coverage percentage relates to the overall top-level category.

Top-level category	*Subcategory*	*Example content*	*% dataset coverage*
Symbolic & Sensory Representations	Sensory details	visual details (colour, illumination, vantage point, shapes, oddities, effects), smell, bodily sensation	15.2%
	Conceptual and	symbolic ideas, cultural and ideological	3.6%

	Metaphorical references	abstractions, metaphorical frames of meaning	
			∑ = 17.8%
Subjective & Personal Experience	Emotions and Affective States	immediate emotional or mood-based affect of the participant	6.8%
	Opinions and Preferences	judgements, likes and dislikes, taste-related stances	6.9%
	Personal Significance	identity-based, relational, symbolic, or spiritual meaning beyond immediate music enjoyment	1.9%
	Personal Time-linked Connections	explicit link to life stage, memory, or other temporally anchored experiences	6.4%
			∑ = 16.6%

Symbolic & Sensory Representations

Symbolic & Sensory Representations encompasses abstract, figurative, and sensory (real or imagined) dimensions of experience, spanning metaphors, values, and perceptual details. *Sensory details* comprised explicit mentions of *visual details* ("lilac", "strobe", "aerial view", "fractals"), *smell* ("incense", "fills the air"), and *bodily sensation* ("tremors", "warm"), whether experienced literally, figuratively, or imaginatively. *Conceptual and Metaphorical references* ("aesthetics", "escapism", "itchy trigger fingers") were grouped with *sensory details* in this theme due their overlap and similarity with *sensory details* in more surreal MIMC descriptions.

Symbolic & Sensory maintained moderate connections to *Narrational* (J = .20) and *Referential* (J = .15) categories, making it the more connected of the two *Experiential* top-level categories. Its strongest links to *Settings* (J = .23), *Actions* (J = .17), *Characters* (J = .16), and *Objects* (J = .15), indicated that symbolic, surreal, or sensory impressions were often anchored to concrete scene elements, attaching figurative or metaphorical descriptions to MIMC. Its modest links to *Chronology* (J = .14) and *Music* (J = .13) reflect participants' occasional use of sensory metaphors to describe musical qualities or to colour time periods and seasonal settings. In the network diagram, *Symbolic & Sensory* sits closer to the main cluster of categories than *Subjective &*

Personal, reflecting its role as a flexible interpretive layer or conceptual colouring that readily attaches to scenes, references, and narrative structures.

Exemplar quotes

- *P648—Classical*: "[...] angelic choirs, peace and relaxation. Very calming and reassuring music accompanying a trustworthy vicar going quietly about his duties in the church. I can smell furniture polish."
- *P695—Electronic*: "gusts of colour, pink, red, whirling upwards, black circles like wily-e-coyote's painted on tunnel bouncing and stretching"
- *P259—Film*: "[...] empty dark space with raindrops falling one by one into large ripples on the floor, open, inspiring, I felt his one in my stomach and it felt great. Open, breathe, fresh, nature"

Subjective & Personal Experience

Subjective & Personal Experience captures how participants reacted to, evaluated, and connected with music in terms of personal feelings, opinions, and meaningful links to people, places, and times in their lives. These included *Emotions and Affective States* ("soothed", "nostalgic"), *Opinions and Preferences* ("awful", "favourite"), *Personal Significance* ("reminds me of her", "things get better"), *Personal Time-linked Connections* ("childhood", "my club days").

Subjective & Personal Experience was the least interconnected category in the coding frame, with its weak internal link to *Symbolic & Sensory* (J = .10) and its consistently low cross-theme co-occurrence (*Narrational* J = .14; *Referential* J = .16). Its strongest links were between *Music* (J = .20) and *Chronology* (J = .14), indicating that personal reactions and memories occasionally intertwined with musical or temporal references, but did not systematically structure MIMC. In the network diagram, *Subjective* sits most peripherally from the main cluster, reflecting how personal, subjective meaning tends to appear as diffuse, individualised overlays rather than a structural component of MIMCs.

Exemplar quotes

- *P559—60s-pop*: "Memories of my mum. Dancing with a beehive haircut - I've not seen her dance to this mus[ic], but I have a picture of her with a beehive in the 60s and imagined she would love dancing to this. Quite sad because it[′]s her birthday today and she has dementia, so this reminded me of my mum as she was, not how she is now.

- *P1338—80s-pop*: "I sang this to my cat when I needed him to turn around so that I could give him his insulin shot. He was sta[n]ding on the dryer, and I had the needle ready. I'm laughing thinking about it."
- *P412—Ambient*: "This brought me to a time when I lived in a different country. Like travelling through time. I[′]m not sure if I am ready for these memories yet. [...]"

When does a thought count? The porous boundary of self-classification between MIMC and no-MIMC reports

All analyses in this study were conducted on the 14,091 valid trials in which participants reported experiencing a thought or scene while listening to the music (MIMC trials). There were, however, 3,750 other valid trials with open-text responses where participants indicated and explained why they did not experience any thoughts or scenes during listening (no-MIMC trials). Importantly, this distinction reflects participants′ explicit self-categorisation of whether their experience constituted a "thought or scene", rather than the presence or absence of mental content. A comparative analysis of MIMC and no-MIMC open-text responses revealed substantial textual overlap, suggesting that classification depended on individual thresholds for what counted as a reportable thought or scene.

For instance, musical recognition and analytic engagement were frequently cited as reasons for no-MIMC responses, with participants stating they were "busy thinking that I had heard this song before and was trying to figure out where" or "concentrating on the music itself and the different instruments". However, near identical responses were often reported as thoughts in MIMC trials, such as "trying to work out initially if [I] knew it" or being "focused mostly on the musical content of this excerpt". Furthermore, fleeting imagery was often explicitly discounted in no-MIMC responses as insufficient to "count", with one participant noting they "briefly saw a man playing a trumpet but that is it, I didn't settle on a scene", while another "dismissed" a thought of "girls dancing on a stage". Conversely, MIMC reports frequently included similarly non-elaborated content as valid, such as "just imagining people playing the instruments, but not a particular scene" or "all I picture is someone with a guitar". Finally, while many no-MIMC trials cited the music being "generic" or "background noise" as the reason for their absence of thoughts, others reported MIMC where “elevator music” or “being on hold” itself constituted their described thoughts. Together, these findings indicate that the boundary between reporting a “thought” or not is not strictly grounded in the presence of mental content, but in listeners’ internally defined criteria for what counts as a reportable thought.

Discussion

From disco floors to deep space, jazz to jellyfish, shadows to spinning colours; working directly with participants’ free-response descriptions—rather than their reduction into computational or quantitative representations—highlights the breadth and complexity of music-influenced mental content (MIMC). Participants’ MIMC descriptions revealed wide phenomenological range, from vivid cinematic narratives to autobiographical memory, abstract visual textures, kinaesthetic simulations, symbolic associations, bodily sensations, and mundane or metacognitive thought.

Beyond documenting this diversity, the present study suggests that MIMCs are best understood as an interconnected system rather than a collection of isolated content types. Our open-ended probing, combined with mixed-method qualitative analyses, revealed patterned relations between content categories that have remained largely uncharted. Our findings expose inter-category dynamics, indicating that imaginative responses to music are organised, scaffolding or colouring others. For example, *Narrational* categories often operated as central hub nodes, linking *Referential* interpretations to *Experiential* subjective meaning. The resulting thematic framework functions as a comprehensive typography of listeners’ music-influenced “thoughtscapes” (van der Walle et al., 2025b), offering a more nuanced tapestry of imagination in action than previously captured by quantitative or computational approaches.

Triadic Thematic Typology

A recurring three-part structure emerged across the dataset: *Narrational*, *Referential*, and *Experiential* themes. Together with the network analysis, these respectively captured the construction of narrative scaffolding, interpretation through cultural reference frames, and shaping through subjective, personal experience.

A clear parallel emerges between our triadic structure and Hashim et al.'s (2023): their "Storytelling" theme aligns most closely with our *Narrational*, "Associations" with *Experiential*, and "References" with *Referential*. Our framework posits that these themes are not merely components of VMI, but reflect structural pillars of any MIMC form and demonstrates how other mental content categories are organised within this structure. By removing the visual bias, we show that these themes operate even when listeners are not necessarily visualising scenes. Our emergent categories also overlap with those identified by Dahl et al. (2022) (e.g., "Nature", "Objects", "Humans", "Film"). Rather than presenting these as a flat list of contents derived from VMI reports using four manipulated songs, our study expands this exploratory framework through a novel hierarchical logic. We demonstrate that these contents extend far beyond VMI—further supported by a broader listener sample and more stylistically diverse music—within a unified MIMC framework.

The prominence of the *Narrational* theme, comprising settings, characters, objects, and actions, fulfills the expectation raised in the introduction that music can shape unfolding mental scenes. The density of co-occurrences with this theme underscores its role as the structural foundation of many MIMCs. Parallels can be observed here to Gabrielsson's (2011) descriptive system for Strong Experiences with Music (SEM) "Cognition: Imagery" category (e.g., landscapes, situations). Often, imagery of landscapes and situations served as a primary cognitive reaction, comparable with the function of our *Narrational* theme, acting as the structural scaffolding upon which other mental content attaches. *Actions* and *Settings* emerged as especially central categories, implying that imagined progression through a scene may provide a core organising node for MIMC. Scenes often became meaningful through what was happening, where it was happening, and how movement unfolded over time. Dahl and colleagues' (2022) categories like "Nature" and "Humans" can be seen forming part of our broader interconnected *Narrational* core, manifesting both in comparable and alternatively fictional, surreal, and referentially rich variants within our unified MIMC framework. Similarly, our network analysis extends Hashim et al.'s (2023) findings by identifying bridge nodes between themes, such as *Actions* linking narrative content to abstracted symbolic representations referentially styled. These bridges reveal a relational architecture not visible in previous content analyses.

The *Referential* theme comprised music, sound, media, art, technology, chronology, and events, unpacking listeners' habitual framing of thoughts through cultural templates—genres, media tropes, aesthetic styles, and time periods. Dahl et al.'s (2022) categories such as "Film" or

"Literal sound", previously treated as imagery sub-types, are re-housed within our *Referential* theme, functioning as interpretive overlays to describe and understand MIMC. This shift moves the focus from content counting what people "see" to functional analysis of how listeners anchor, interpret, and elaborate on their various MIMC manifestations. *Referential* content frequently layered onto *Narrational* components, framing scenes as cinematic, historical, genre-specific, futuristic, or otherwise recognisable. Gabrielsson (2011) describes a comparable SEM category, "Musical perception-cognition", where music acts as a "time machine" into cultural heritage. Furthermore, participants' use of *Media* and *Chronology* references to anchor their MIMC in recognisable worlds ("like a 40s horror film", "a Renaissance court", "a retro game level") resonates with Thompson et al.'s (2023) concept of "source sensitivity"—the capacity to track the causes, contexts, and cultural origins associated with music. Although Thompson et al. apply this concept to music appreciation, our findings indicate that the same cognitive orientation shapes listeners' thought content. By mapping MIMC onto familiar cultural sources, *Referential* content provides the interpretive grammar through which listeners make sense of their mental content. Imagination is not free-floating but culturally situated, shaped by the artworlds, genres, and narrative conventions listeners have internalised. Identifying *Referential* as a major theme therefore highlights MIMC as not only perceptual or emotional but also deeply cultural, drawing on shared symbolic resources to construct meaning.

The *Experiential* theme addresses the emotional appraisals and sensory experiences of MIMC raised in the introduction, capturing symbolic, abstract, and sensory representations alongside subjective experiences like emotional reactions, preferences, and personal significance. These MIMC aspects were the rarest and most intimate, with one participant writing their "feeling[s] are too deeply personal and connects me to my past". This echoes Gabrielsson's (2011) observation that strong experiences often touch one's "innermost self". The structural isolation of *Subjective & Personal experience* in our network analysis reinforces this; while personal meaning is not a central organising force of MIMC, it is a delicate overlay that emerges selectively. The domains of self-oriented appreciation from Thompson et al. (2023) also map onto this theme, showing how similar self-related processes shape listeners' MIMCs: visceral sensations in *Symbolic & Sensory* reflect the "ecological self", reinforcement through bodily feedback; *Personal time-linked connections* reflect the "extended self", capturing memory and personal identity; and preferences and emotional reactions and map onto the "private self", resonating with inner feelings and self-understanding.

Crucially, the principal contribution of the present study is not only the identification of themes, but evidence that these themes interact systematically. Previous qualitative studies have catalogued the contents of musical imagination, but our findings map the relations between contents and argue for their equal importance in understanding MIMC. For instance, a listener may imagine a person running through neon-lined streets (*Narrational*), interpret it as resembling a cyberpunk film (*Referential*), and experience excitement, empowerment, or anemoia (*Experiential*). These are not separate responses occurring independently, but interacting components co-assembling into a single, coherent imaginative episode. Together, these themes reveal that MIMCs are not singular phenomena but a dynamic system—a coordinated interplay between narrative scaffolding, cultural reference frames, and subjective meaning-making. This systemic view opens new avenues for theorising imagination as a distributed, multidimensional process shaped by both internal and external sources.

Methodological implications

The present findings reinforce concerns that imaginative experiences are frequently flattened by quantitative scales, predefined response categories, and methods that isolate one target phenomenon at a time. Hashim et al. (2023), in their isolated examination of VMI, noted that participants' VMI descriptions often contained non-visual elements like affect or musical features, but were not acknowledged as intrinsic elements of such inner experiences. When participants are asked only about imagery, memories, or emotions, broader thoughtscapes may remain hidden. Our framework addresses this omission by recognising all mental content as integral, foundational components of MIMC. By validating and elevating previously unaccounted for MIMC forms within top-level categories (e.g., *Music & Sounds*/*Symbolic & Sensory* within *Experiential*/*Referential* themes), we demonstrate that they operate alongside other MIMC, such as visual content, rather than being subordinate to it.

Open-response methods preserve ambiguity, hybridity, and overlap between mental content that more restrictive methods may miss. Moreover, human-driven qualitative enabled identification of the finer connective tissue between imaginative experiences that computational and purely quantitative approaches often overlook. In the intricate case of imagination, such approaches are most powerful when grounded in rich descriptive data paired with interpretive analysis. Mixed-method approaches may therefore offer the strongest route forward: qualitative methods to preserve nuance, and quantitative methods to model structure at scale.

Future research directions

Future studies could formalise the relational architecture observed here treating MIMC as a networked cognitive system rather than a set of isolated components. Dynamical-systems modelling could capture how imaginative landscapes evolve moment-to-moment during listening, offering a temporal map of how scenes, references, and affect emerge, stabilise, and dissolve. Further network-analytic approaches could model how thought types co-occur across time, revealing whether certain categories reliably precede or scaffold others (e.g., whether *Narrational* content "sets the stage" for *Experiential* MIMC) and under what conditions these relation shifts. For example, the relatively peripheral position of *Subjective & Personal experience* may shift when listeners engage with self-selected or personally significant music, where autobiographical and self-relevant content could become more central. Such approaches would move MIMC research toward a mechanistic account of imagination, capturing not only what people think, but how their thoughtscapes unfold as dynamic, interdependent processes.

Refining theoretical models of MIMC also requires attention to individual differences and contextual situations. As MIMC appear to be shaped by both musical features and cultural templates (Ayyildiz et al., 2025; Margulis et al., 2022a; Jakubowski & Francini, 2023; Margulis et al., 2022b), further research could examine how cultural background, listening context, and an even wider range of musical genres structurally mould thoughtscapes. Cross-cultural studies could reveal universal or cultural specificity of *Referential* categories (e.g., media tropes, historical periods, genre labels). Likewise, manipulating listening contexts—solitary vs social, focused vs distracted—could illuminate how situational factors modulate the balance between *Narrational*, *Referential*, and *Experiential* content (e.g., Herff et al., 2025). Such work could clarify the extent to which different MIMC forms are driven by musical structure, personal history, cultural knowledge, and their interaction. Longitudinal or experience-sampling approaches could further track how stable these imaginative signatures are within individuals, and how they vary with mood, context, or repeated exposure.

Conclusion

MIMC appears less like a set of separate phenomena and more like a coordinated ecology of thought. Music can prompt listeners to simulate scenes, draw on cultural knowledge, and invoke personal meaning in mutually reinforcing ways. By foregrounding the interconnected structure of these experiences, the present study offers a broader account of imagination in music listening—one in which narratives, references, symbolism, and evaluations do not compete for explanation, but combine to form the dynamic thoughtscapes of everyday listening. Our findings also demonstrate the value of a more open approach to MIMC research, inclusive of the full spectrum of listeners' imaginative experiences. By adopting this stance, we lay the groundwork for future MIMC models that treat thought as an interconnected system, better equipped to capture the nuance and diversity of music-shaped thoughtscapes.

Our findings also highlight a deeper methodological complication: the difficulty of deciding where a "thought" begins. Many free-responses reported as no-MIMC trials contained descriptions nearly indistinguishable from those recorded as "thoughts" (e.g., recognising melodies, analysing instrumentation, impressions of mood or genre). The critical difference was not the presence or absence of internal experience, but rather how participants classified that experience. Fleeting imagery, generic impressions, and momentary associations were dismissed by some as too insubstantial to "count", whereas others offered equivalent content as valid thought descriptions.

In this light, the boundary between a "thought" and a "no thought" category appears porous and individually negotiated rather than objectively defined. An open, participant-led approach can make this visible by showing what counts as a thought is not simply detected, but actively interpreted by the listener in the moment. The act of self-report requires listeners to draw a line that the data suggests is inherently ambiguous: a thought does not simply occur or fail to occur, but is brought to report through the criteria a listener applies. The issue, then, is not only whether the internal experience exists, but whether it is judged as something worth naming. This ambiguity underscores an important theoretical challenge for studying MIMC directly: the more we ask participants to notice their inner world, the more they must interpret, judge, and categorise it, potentially shifting the very phenomenon we aim to capture. Future research must account for this subjectivity—not as noise to be eliminated, but as a fundamental feature—in order to get closer to understanding the human thoughtscape.

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